

BetaGo Mirco

A summarized plan of the planned bachelor thesis

Idea

- Go on 9x9 board
 - Retrain AlphaGo Zero to have comparable training background
 - Goal: A capable Go AI
1. A difficulty estimator -> Is it possible to reduce simulation cost by relying on a difficulty estimator that dynamically limits the maximal depth of the MCTS and this limit the necessary amount of calculations?
 - Maybe additional ways of limiting MCTS search space
 2. (Or teach a model the rules/make it represent the learned rules -> How does the AI understand the rules it has learned? - A teaching AI that can symbolize concepts)

Experiments

Metrics

in regards to

- win rate
 - number of simulations/calculations
1. Compare single net combining diff. est. and policy network and separate nets for diff. est and policy
 2. Play against KataGo and compare in metrics
 3. Use pretrained KataGo as evaluator on different game phases for accuracy

Structure

1. Introduce the topic and the thesis
2. Explain history of Go AI, and important concepts (MCTS)
 - For the teacher model, also briefly explain the rules of Go
3. (Explain the base model, e.g. AlphaGo, AlphaZero and) Explain the model
4. Experiments
5. Conclusion

Environment

- Environment: Google Collab
- Libraries: PyTorch, OpenSpiel (as the basis for the game)

Timeline

- 3 Weeks: Development and training of models
- Month 2: Experiments
- Month 3: Evaluation, writing, finalization